

A sharp universal profile for multivariate characteristic-function factorization

Alexandr Martinevski

Independent Researcher, Lithuania

alex.martinevski@gmail.com

ORCID: 0009-0000-4230-4414

September 2026

AI assistance disclosure. AI systems used in the research and manuscript workflow included OpenAI GPT-5.6 Sol and Anthropic Claude Opus 5.5. They were used for mathematical exploration, proof development and critique, symbolic and numerical verification, literature search, and manuscript drafting and editing. AI systems are not authors. The author reviewed the mathematical statements and proofs and takes full responsibility for the content.

Abstract

For complex random variables $Z_1, \dots, Z_d$ with $|Z_j| \leq 1$, prescribe the moduli $r_j = |\mathbb{E}Z_j|$, put $\theta_j = \arccos r_j$, $R = \prod_j r_j$ and $s = \sum_j \theta_j$. We prove the exact profile

$$\sup \left\{ \left| \mathbb{E} \prod_j Z_j - \prod_j \mathbb{E} Z_j \right| : |Z_j| \leq 1, |\mathbb{E} Z_j| = r_j \right\} = R - \cos(\min\{s, \pi\}).$$

The sharp universal characteristic-function factorization constant is therefore the one-dimensional corollary

$$C_d = \max_{0 \leq x \leq \pi} \{\cos^d(x/d) - \cos x\} = \max_{0 \leq \theta \leq \pi/d} \{\cos^d \theta - \cos(d\theta)\} = \|T_d - x^d\|_{\infty, [-1, 1]}.$$

It is attained by a common fair sign and is simultaneously sharp for mutual-independence, sub-independence and one-variable characteristic-function defects. The profile yields a short proof based only on Cauchy–Schwarz and log-concavity of cosine. We also record selected consequences of the real-parameter interpolation $C(p) = \max_{0 \leq \theta \leq \pi/p} \{\cos^p \theta - \cos(p\theta)\}$: $C(p)$ is strictly increasing for $p > 2$, every maximizer lies in $(\pi/(p+1), \pi/p)$, and the classical point $\pi/(p+1)$ is asymptotically sharp. In particular,

$$C_d - \left\{ \cos \frac{\pi}{d+1} + \cos^d \frac{\pi}{d+1} \right\} = \frac{\pi^6}{8d^4} + O(d^{-5}),$$

which gives a four-term expansion of C_d . We retain exact real-valued and Gaussian comparison results and an algebraic description of the critical point.

1 Introduction

Mutual independence of random blocks $X_j \in \mathbb{R}^{p_j}$ is equivalent to the factorization $\varphi_X(t_1, \dots, t_d) = \prod_j \varphi_{X_j}(t_j)$. Empirical-characteristic-function tests of total independence go back at least to [4, 10]; related L^2 constructions include distance multivariate and kernel criteria [2, 16]. For two blocks, Daniušis et al. introduced the uniform Fourier dependence measure (UFDM), the L^∞ norm of the joint-minus-product characteristic function, and proved

the universal bound one [6]. The weaker diagonal factorization property is sub-independence [7, 8, 18].

The present line of work has an earlier origin. In an unpublished student research project of the author (1996), the uniform characteristic-function factorization defect was introduced as a dependence quantity, with particular motivation from cryptography; the limiting value 2 and Gaussian calculations were already part of that project. The exact finite- d profile, the sharp constants and all proofs in the present paper are new developments from 2026.

The first question here is not estimation but the population extremal problem: how large can the uniform factorization defect be for d blocks? A previous formulation of this paper answered this by a direct angular proof. The stronger result below gives the entire sharp profile after the marginal mean moduli are fixed. The universal constant C_d then follows in a few lines. This reorganization has two advantages: the proof is shorter, and the geometry of the extremizer is visible before the final one-dimensional maximization.

Our second aim is to record only those consequences of the continuous interpolation

$$C(p) = \max_{0 \leq \theta \leq \pi/p} \{\cos^p \theta - \cos(p\theta)\}, \quad p > 2,$$

that sharpen the original sequence $C_d = C(d)$. The broader two-parameter cosine/Chebyshev surface and its scaling limits are left for separate work. In particular, we use the interpolation to obtain continuous monotonicity and a sharp asymptotic correction to the elementary point $\pi/(d+1)$.

The fixed-marginal problem is treated separately in [14]; in particular, that companion work gives exact results for several symmetric unimodal marginals, a general large- d “price of marginals”, and a non-symmetric extension. The local arithmetic, irreducibility and Galois theory of the critical polynomial introduced below are developed in [15].

The identity $C_d = \|T_d - x^d\|_\infty$ should not be confused with the classical Chebyshev minimax problem for monic polynomials, where the optimally scaled polynomial is $2^{1-d}T_d$ [17]. Our searches did not locate the profile theorem below, the sharp d -linear complex unit-ball constant, or the sequence C_d for $d \geq 3$; this is a statement about the searches performed, not a claim of priority. Abstract Grüss and multiplicativity-defect inequalities form relevant background, but they do not by themselves give the constant below.

2 Exact fixed-mean profile and the universal constant

For $r = (r_1, \dots, r_d) \in [0, 1]^d$ define

$$\mathcal{C}_d(r_1, \dots, r_d) = \sup \left\{ \left| \mathbb{E} \prod_{j=1}^d Z_j - \prod_{j=1}^d \mathbb{E} Z_j \right| : |Z_j| \leq 1, |\mathbb{E} Z_j| = r_j \right\}.$$

Put

$$\theta_j = \arccos r_j, \quad R = \prod_{j=1}^d r_j, \quad s = \sum_{j=1}^d \theta_j.$$

Theorem 2.1 (Exact profile). *For every $d \geq 1$ and $r \in [0, 1]^d$,*

$$\boxed{\mathcal{C}_d(r_1, \dots, r_d) = R - \cos(\min\{s, \pi\}).}$$

If $s \leq \pi$, equality is attained by $Z_j = e^{i\varepsilon\theta_j}$ with a common fair sign $\mathbb{P}(\varepsilon = \pm 1) = 1/2$. If $s \geq \pi$, equality is attained by a coupling with $\prod_j Z_j \equiv -1$ and the prescribed means.

For random blocks $X = (X_1, \dots, X_d)$ define

$$\Delta_X(t) = \varphi_X(t_1, \dots, t_d) - \prod_{j=1}^d \varphi_{X_j}(t_j), \quad D_d(X) = \sup_t |\Delta_X(t)|,$$

and let D_d^* be the supremum over all joint laws and all block dimensions.

Corollary 2.2 (Sharp universal constant). *For every $d \geq 2$,*

$$D_d^* = C_d := \max_{0 \leq x \leq \pi} \{\cos^d(x/d) - \cos x\} = \max_{0 \leq \theta \leq \pi/d} \{\cos^d \theta - \cos(d\theta)\} = \|T_d - x^d\|_{L^\infty[-1,1]}.$$

Equivalently,

$$\left| \mathbb{E} \prod_j Z_j - \prod_j \mathbb{E} Z_j \right| \leq C_d \quad (|Z_j| \leq 1),$$

and the bound is sharp. It is attained by $Z_j = e^{i\varepsilon\theta_d^*}$, or by $X_1 = \dots = X_d = \varepsilon$ at the common frequency θ_d^* .

Corollary 2.3. *The same constant is sharp for the sub-independence defect*

$$D_d^{\text{sub}}(X) = \sup_t \left| \varphi_{X_1+\dots+X_d}(t) - \prod_j \varphi_{X_j}(t) \right|,$$

for the one-variable defect $S_d(X) = \sup_t |\varphi_X(dt) - \varphi_X(t)^d|$, and after restricting D_d to exchangeable laws or to laws with identical marginals.

Theorem 2.4. *For $d \geq 3$, $g_d(\theta) = \cos^d \theta - \cos(d\theta)$ has a unique maximizer $\theta_d^* \in (0, \pi/d)$. With $c_d^* = \cos \theta_d^*$,*

$$U_{d-1}(c_d^*) = (c_d^*)^{d-1}.$$

The maximizers of $|T_d(x) - x^d|$ on $[-1, 1]$ are exactly $\pm c_d^$. Moreover $C_2 < C_3 < C_4 < \dots$ and $C_d \uparrow 2$.*

3 Proof of the profile theorem and Corollary 2.2

Proof of Theorem 2.1. Multiplying coordinates by deterministic unimodular constants, assume $\mathbb{E} Z_j = r_j \geq 0$. We argue by induction on d ; the case $d = 1$ is trivial. Put

$$W = \prod_{j < d} Z_j, \quad u = \mathbb{E} W, \quad R' = \prod_{j < d} r_j, \quad s' = \sum_{j < d} \theta_j, \quad r = r_d = \cos \theta, \quad \theta = \theta_d.$$

Then

$$\mathbb{E}(W Z_d) - r R' = \mathbb{E}[(W - u)(Z_d - r)] + r(u - R'),$$

and Cauchy–Schwarz gives

$$|\Delta| \leq \sin \theta \sqrt{1 - |u|^2} + r t, \quad t := |u - R'|, \quad (1)$$

because $\mathbb{E}|Z_d - r|^2 \leq 1 - r^2$ and $\mathbb{E}|W - u|^2 \leq 1 - |u|^2$.

By induction,

$$0 \leq t \leq R' - \cos(\min\{s', \pi\}).$$

The reverse triangle inequality gives $|u| \geq |R' - t|$. Write $R' - t = \cos \phi$ with

$$\phi \in [\arccos R', \min\{s', \pi\}].$$

Then $\sqrt{1 - |u|^2} \leq \sin \phi$, and (1) yields

$$|\Delta| \leq \sin \theta \sin \phi + \cos \theta (R' - \cos \phi) = R - \cos(\theta + \phi).$$

If $s \leq \pi$, then $\theta + \phi \leq s$ and the right-hand side is at most $R - \cos s$. If $s \geq \pi$, the admissible interval contains $\phi = \pi - \theta$ (since $\arccos R' \leq \pi/2 \leq \pi - \theta$ and $s' \geq \pi - \theta$), so the upper bound is $R + 1$. This proves the required upper bound.

If $s \leq \pi$, take a common fair sign ε and $Z_j = e^{i\varepsilon\theta_j}$. Then $\mathbb{E}Z_j = r_j$ and $\mathbb{E}\prod_j Z_j = \cos s$, giving $R - \cos s$.

Suppose $s \geq \pi$ and set

$$Q = \left\{ r \in [0, 1]^d : \sum_j \arccos r_j \geq \pi \right\}.$$

The set Q is compact and convex because $\arccos$ is concave. Every extreme point is either on the boundary $\sum_j \arccos r_j = \pi$ or is a cube vertex: if the sum is strictly larger than π and some coordinate lies in $(0, 1)$, a sufficiently small perturbation of that coordinate in both directions stays in Q . A cube vertex belongs to Q exactly when it has at least two zero coordinates. Boundary points are realized by the common-sign construction with $\sum_j \theta_j = \pi$, for which $\prod_j Z_j = -1$. A vertex with $k \geq 2$ zero coordinates is realized by taking the remaining coordinates equal to 1, choosing $U_1, \dots, U_{k-1}$ independently uniform on the unit circle, and setting $U_k = -\overline{U_1 \cdots U_{k-1}}$; then every zero-coordinate mean is 0 and the product is identically -1 . By the finite-dimensional Minkowski theorem, every $r \in Q$ is a convex combination of extreme points; mixing the corresponding couplings preserves the means and keeps the product identically -1 . Hence the value $1 + R$ is attained. $\square$

Proof of Corollary 2.2. Let $\theta_j = \arccos r_j$. If $s = \sum_j \theta_j \leq \pi$, strict concavity of $\log \cos$ gives

$$R = \prod_j \cos \theta_j \leq \cos^d(s/d),$$

with equality when the angles are equal. Thus Theorem 2.1 gives

$$\mathcal{C}_d(r) \leq \cos^d(s/d) - \cos s.$$

If $s \geq \pi$, the same concavity and monotonicity give $R \leq \cos^d(s/d) \leq \cos^d(\pi/d)$, so $\mathcal{C}_d(r) = 1 + R \leq 1 + \cos^d(\pi/d)$, the endpoint value at $s = \pi$. Therefore

$$\sup_r \mathcal{C}_d(r) = \max_{0 \leq s \leq \pi} \{\cos^d(s/d) - \cos s\} = C_d.$$

Equal angles and the common fair sign attain every value in this one-dimensional family, hence in particular its maximum. Taking $Z_j = e^{i\langle t_j, X_j \rangle}$ proves the characteristic-function bound, while scalar angle-valued variables realize the extremizer.

Finally, for $x \in [-1, 1]$, parity reduces $|T_d(x) - x^d|$ to $x \geq 0$; writing $x = \cos \theta$ gives the defect of the common-sign construction, so it is at most C_d . Equality occurs at $x = \cos \theta_d^*$. $\square$

Proof of Corollary 2.3. Each restricted problem is bounded by the full one. For a common Rademacher sign ε , $\varphi_\varepsilon(d\theta) = \cos(d\theta)$ and $\varphi_\varepsilon(\theta)^d = \cos^d \theta$, so the one-variable problem already attains C_d ; the same law is exchangeable and has identical marginals. $\square$

4 The optimizer and a continuous interpolation

Proof of Theorem 2.4. Since $\sin d\theta = \sin \theta U_{d-1}(\cos \theta)$,

$$g'_d(\theta) = d \sin \theta [U_{d-1}(\cos \theta) - \cos^{d-1} \theta].$$

Using

$$\frac{U_{d-1}(x)}{x^{d-1}} = 2^{d-1} \prod_{k=1}^{\lfloor (d-1)/2 \rfloor} \left(1 - \frac{\cos^2(k\pi/d)}{x^2} \right), \quad x > 0,$$

(with the usual central factor when d is even), the ratio is strictly increasing on $(\cos(\pi/d), 1]$, from 0 to $d > 1$. Hence it crosses 1 exactly once, proving uniqueness. The assertion about $|T_d - x^d|$ follows from Corollary 2.2 and parity.

Strict growth of C_d follows from Proposition 4.1 below. Finally $C_d \geq 1 + \cos^d(\pi/d) \rightarrow 2$ and $C_d \leq 2$. $\square$

For real $p > 2$ define

$$C(p) = \max_{0 \leq \theta \leq \pi/p} \{\cos^p \theta - \cos(p\theta)\} = \max_{0 \leq s \leq \pi} \{\cos^p(s/p) - \cos s\}.$$

Thus $C(d) = C_d$.

Proposition 4.1 (Selected continuous consequences). *For $p > 2$:*

1. $C(p)$ is strictly increasing in p ;
2. every maximizer θ_p is interior and satisfies

$$\frac{\pi}{p+1} < \theta_p < \frac{\pi}{p}, \quad \sin(p\theta_p) = \sin \theta_p \cos^{p-1} \theta_p;$$

3. with $a_p = \pi/(p+1)$,

$$\theta_p = a_p + \frac{\pi^3}{2p^3} + O(p^{-4}), \quad C(p) - \{\cos^p a_p + \cos a_p\} = \frac{\pi^6}{8p^4} + O(p^{-5}).$$

Proof. For fixed $s \in (0, \pi]$,

$$\frac{\partial}{\partial p} [p \log \cos(s/p)] = \log \cos z + z \tan z > 0, \quad z = s/p \in (0, \pi/2),$$

because the derivative of $\log \cos z + z \tan z$ is $z \sec^2 z > 0$ and the expression vanishes at $z = 0$. Thus $\cos^p(s/p)$, and hence its maximum after subtracting $\cos s$, increases strictly with p .

At an endpoint $\theta = \pi/p$ the derivative is negative, whereas the objective is positive for small $\theta > 0$; therefore a maximizer is interior and satisfies the displayed stationary equation. Put $\delta = \pi - p\theta_p$. Then $\sin \delta = \sin \theta_p \cos^{p-1} \theta_p < \sin \theta_p$. Since $0 < \theta_p < \pi/p < \pi/2$ and $\delta < \pi - \theta_p$, this forces $\delta < \theta_p$, hence $\theta_p > \pi/(p+1)$.

For the last assertion, expand the stationarity equation around a_p . One obtains $f'_p(a_p) = \pi^3/(2p) + O(p^{-2})$ and $f''_p(a_p) = -p^2 + O(p)$ for $f_p(\theta) = \cos^p \theta - \cos(p\theta)$, hence $\theta_p - a_p = -f'_p(a_p)/f''_p(a_p) + O(p^{-4}) = \pi^3/(2p^3) + O(p^{-4})$. Taylor expansion at the maximizer then gives the value correction $\pi^6/(8p^4) + O(p^{-5})$. $\square$

Remark 4.2. The full continuous theory also gives a Gamma/Fourier realization and a two-parameter cosine/Chebyshev master surface. Those results are deliberately not imported here: Proposition 4.1 contains only the pieces that sharpen the original sequence C_d .

5 Algebraic description and small dimensions

Corollary 5.1. $C_2 = 1$, $C_3 = 2/\sqrt{3}$, $C_4 = 9/7$.

Proof. For $d = 2$, $g_2(\theta) = \sin^2 \theta$. For $d = 3$, $x^3 - T_3(x) = 3x(1 - x^2)$, maximal at $x = 1/\sqrt{3}$. For $d = 4$, with $y = \cos^2 \theta$, $g_4 = -7y^2 + 8y - 1$, maximal at $y = 4/7$. $\square$

Proposition 5.2. For $d \geq 3$, $c_d^* = \cos \theta_d^*$ is algebraic and

$$C_d = U_{d-2}(c_d^*).$$

Moreover $y = (c_d^*)^2$ satisfies an integer polynomial of degree $\lfloor (d-1)/2 \rfloor$ with leading coefficient $2^{d-1} - 1$. In particular

$$C_5 = \frac{4}{15}(2\sqrt{21} - 3)\sqrt{\frac{6+\sqrt{21}}{15}}, \quad C_6 = \frac{225 + 140\sqrt{70}}{961}.$$

Proof. The critical equation is $U_{d-1}(c) = c^{d-1}$. For odd d it is even in c ; for even d both sides are odd and we divide by c , giving the stated polynomial in $y = c^2$. Finally $T_d(c) = cU_{d-1}(c) - U_{d-2}(c) = c^d - U_{d-2}(c)$ at a critical point, so $C_d = c^d - T_d(c) = U_{d-2}(c)$. $\square$

Remark 5.3 (The tangent-square critical polynomial). Writing $x = \tan^2 \theta$ in the critical equation $\sin(d\theta) = \sin \theta \cos^{d-1} \theta$ gives

$$A_d(x) := \sum_{k=0}^{\lfloor (d-1)/2 \rfloor} (-1)^k \binom{d}{2k+1} x^k - 1 = 0.$$

Thus A_d is not introduced ad hoc: its distinguished positive root is exactly $x_d^* = \tan^2 \theta_d^*$, where θ_d^* is the optimizer defining C_d . The arithmetic of this family—irreducibility, local Newton polygons, ramification and Galois groups—is developed in [15]. In particular, that work shows that the algebraic problem has substantial structure well beyond the small-dimensional formulas recorded here.

6 The real case

Theorem 6.1. For real random variables with $|X_j| \leq 1$,

$$\sup \left| \mathbb{E} \prod_j X_j - \prod_j \mathbb{E} X_j \right| = 1 + \left(1 - \frac{2}{d}\right)^d,$$

attained when exactly one coordinate equals -1 , at a uniformly random position, and the others equal 1. The value increases to $1 + e^{-2} \approx 1.1353$, which is strictly below C_d for every $d \geq 3$.

Proof. Randomizing each X_j to ± 1 with conditional mean X_j reduces to $U_j \in \{\pm 1\}$. Put $A = \mathbb{E} \prod U_j$, $a_j = \mathbb{E} U_j$, $B = \prod a_j$. If $AB \geq 0$ then $|A - B| \leq 1$. Otherwise flipping U_1 changes (A, B) to $(-A, -B)$, so we may assume $B > 0 > A$; flipping negative a_j in pairs we may assume all $a_j > 0$. With $q_j = \mathbb{P}(U_j = -1) < \frac{1}{2}$, $N = \#\{j : U_j = -1\}$ and $\pi = \mathbb{P}(N \text{ even})$ we have $A = 2\pi - 1$ and $\sum_j q_j = \mathbb{E} N \geq 1 - \pi$. By AM–GM, $B = \prod (1 - 2q_j) \leq (1 - \frac{2}{d} \sum q_j)^d \leq (1 - \frac{2(1-\pi)}{d})^d$, so $B - A \leq 1 + \psi(\pi)$ with $\psi(\pi) = (1 - 2(1 - \pi)/d)^d - 2\pi$ non-increasing on $[0, \frac{1}{2}]$; hence $B - A \leq 1 + (1 - 2/d)^d$. The extremal law has $A = -1$, $a_j = 1 - 2/d$. $\square$

Remark 6.2. For nonnegative factors a classical lemma of Ibragimov [9] controls the same functional under strong mixing. Deviations from the piling-up lemma of linear cryptanalysis [12] under dependence were studied by Kukorelly [11]; Theorem 6.1 gives the sharp worst case for ± 1 -valued variables.

7 Gaussian laws

Proposition 7.1. *If $X = (X_1, \dots, X_d)$ is Gaussian, then $D_d(X) \leq 1$, with strict inequality when the joint covariance matrix is nonsingular. The value 1 is attained (as a supremum over t) by singular Gaussian laws, e.g. $X_1 = Z$, $X_2 = -Z$ with $Z \sim N(0, \sigma^2)$, for which $D_2 = \sup_t (1 - e^{-t^2 \sigma^2}) = 1$.*

Proof. With mean μ , joint covariance Σ and block covariances Σ_{jj} , after removing the common factor $e^{i\langle t, \mu \rangle}$ both terms are real numbers in $(0, 1]$: $|\Delta_X(t)| = |e^{-t^\top \Sigma t/2} - e^{-\sum_j t_j^\top \Sigma_{jj} t_j/2}| < 1$. If Σ is nonsingular both exponents tend to $+\infty$ together as $\|t\| \rightarrow \infty$ and stay bounded together on bounded sets, so the supremum is attained and is strictly below one. $\square$

For two blocks the defect is explicit. Let $h(\lambda) = |\lambda - 1| \lambda^{-\lambda/(\lambda-1)}$ for $\lambda > 0$, $\lambda \neq 1$, and $h(1) = 0$.

Proposition 7.2 (explicit form of the supremum in [6, Thm. 1(4)]). *Let (X, Y) be jointly Gaussian with nonsingular block covariances and largest canonical correlation $\rho_1 \in [0, 1)$. Then*

$$D_2(X, Y) = h(1 - \rho_1) = \rho_1(1 - \rho_1)^{(1-\rho_1)/\rho_1} \quad (\rho_1 > 0),$$

and $D_2 = 0$ if $\rho_1 = 0$. In particular $D_2 \sim \rho_1/e$ as $\rho_1 \downarrow 0$ and $D_2 \rightarrow 1$ as $\rho_1 \uparrow 1$.

Proof. Blockwise whitening does not change D_2 and reduces the joint covariance to $R = \begin{pmatrix} I & C \\ C^\top & I \end{pmatrix}$, whose eigenvalues lie in $[1 - \rho_1, 1 + \rho_1]$ with both endpoints attained. Writing $t = ru$ with $\|u\| = 1$, $\lambda = u^\top R u$ and $x = r^2/2$, $|\Delta(t)| = |e^{-\lambda x} - e^{-x}|$. For fixed $\lambda \neq 1$ the maximum over $x \geq 0$ is attained at $x = \log \lambda / (\lambda - 1)$ and equals $h(\lambda)$. One checks $h(\lambda) = h(1/\lambda)$, and h increases on $[1, \infty)$ because $e^{-x} - e^{-\lambda x}$ increases in λ for each $x > 0$. Since $(1 - \rho_1)(1 + \rho_1) < 1$, i.e. $1/(1 - \rho_1) > 1 + \rho_1$, the maximum over λ is $h(1 - \rho_1)$. $\square$

Remark 7.3. The same argument gives, for any two Gaussian laws with a common mean and positive definite covariances A, B , $\sup_t |\varphi_{N(\mu, A)}(t) - \varphi_{N(\mu, B)}(t)| = \max\{h(\lambda_{\min}), h(\lambda_{\max})\}$, where $\lambda_{\min}, \lambda_{\max}$ are the extreme generalized eigenvalues of $Av = \lambda Bv$; equivalently it is an increasing function of the Thompson distance $\|\log(B^{-1/2}AB^{-1/2})\|_{\text{op}}$.

8 Large- d behaviour

Let

$$\tilde{C}_d = \cos \frac{\pi}{d+1} + \cos^d \frac{\pi}{d+1}.$$

The elementary point $\theta = \pi/(d+1)$ already gives $\tilde{C}_d \leq C_d$. We first record a uniform finite- d estimate and then the sharper asymptotic correction.

Proposition 8.1 (Global approximant). *For every $d \geq 4$,*

$$0 \leq C_d - \tilde{C}_d \leq \frac{\pi^6}{4d^4}.$$

Moreover, with $\delta_d^* = \pi - d\theta_d^*$,

$$\left| \delta_d^* - \frac{\pi}{d+1} \right| \leq \frac{\pi^3}{2d^2}.$$

Proof. For $\delta \in [0, \pi]$ put

$$x(\delta) = \frac{\pi - \delta}{d}, \quad G(\delta) = \cos^d x(\delta) + \cos \delta.$$

Then $C_d = \max_{[0, \pi]} G$. On $[\pi/2, \pi]$, $G \leq 1 < 1 + \cos^d(\pi/d) = G(0)$, while

$$G'(\delta) = \sin x \cos^{d-1} x - \sin \delta.$$

Thus every maximizer $\delta_* \in (0, \pi/2)$ satisfies $\sin \delta_* = \sin x_* \cos^{d-1} x_*$, where $x_* = x(\delta_*) \leq \pi/d \leq \pi/4$.

From $\sin \delta_* \leq x_* \leq \pi/d$ and $\sin \delta \geq 2\delta/\pi$ on $[0, \pi/2]$, $\delta_* \leq \pi^2/(2d)$. Also $\sin \delta_* \geq \delta_* - \delta_*^3/6$ gives

$$\delta_* \leq \frac{\pi}{d+1} + \frac{\pi^6}{48d^3}.$$

For the lower bound, Bernoulli's inequality and $x_* \leq \pi/4$ give $\cos^{d-1} x_* \geq 1 - (d-1)x_*^2/2 \geq 0$, and hence

$$\delta_* \geq \sin \delta_* \geq \left(x_* - \frac{x_*^3}{6}\right) \left(1 - \frac{(d-1)x_*^2}{2}\right) \geq x_* - \frac{d}{2}x_*^3 \geq \frac{\pi - \delta_*}{d} - \frac{\pi^3}{2d^2}.$$

Therefore

$$\left|\delta_* - \frac{\pi}{d+1}\right| \leq \frac{\pi^3}{2d^2}.$$

Finally

$$G''(\delta) = -\frac{1}{d} \left[\cos^d x - (d-1) \sin^2 x \cos^{d-2} x \right] - \cos \delta,$$

and $|G''| \leq 2$ for $d \geq 4$. Taylor's formula at δ_* therefore yields

$$G\left(\frac{\pi}{d+1}\right) \geq C_d - \left(\delta_* - \frac{\pi}{d+1}\right)^2 \geq C_d - \frac{\pi^6}{4d^4}.$$

Since $G(\pi/(d+1)) = \tilde{C}_d \leq C_d$, the proof is complete. $\square$

Proposition 8.2 (Sharp asymptotic correction). *As $d \rightarrow \infty$,*

$$\theta_d^* = \frac{\pi}{d+1} + \frac{\pi^3}{2d^3} + O(d^{-4}), \quad \boxed{C_d - \tilde{C}_d = \frac{\pi^6}{8d^4} + O(d^{-5})}.$$

Thus Proposition 8.1 is sharp in order.

Proof. This is Proposition 4.1(3) with $p = d$. $\square$

Corollary 8.3 (Four-term expansion). *As $d \rightarrow \infty$,*

$$\begin{aligned} C_d = 2 - \frac{\pi^2}{2d} + \frac{\pi^2(\pi^2 + 4)}{8d^2} - \frac{\pi^2(\pi^4 + 28\pi^2 + 24)}{48d^3} \\ + \frac{\pi^2(\pi^6 + 112\pi^4 + 624\pi^2 + 192)}{384d^4} + O(d^{-5}). \end{aligned}$$

Consequently

$$C_{d+1} - C_d = \frac{\pi^2}{2d^2} - \frac{\pi^2(\pi^2 + 6)}{4d^3} + O(d^{-4}),$$

and C_d is strictly concave for all sufficiently large d .

Remark 8.4. Numerically the discrete concavity appears to hold already from $d = 2$; a proof for every d remains open.

9 Remarks

Operator and dynamical forms. Since $C_d = \|T_d - x^d\|_{\infty, [-1, 1]}$ and the maximizers are $\pm c_d^*$ (Theorem 2.4), the spectral theorem gives $\|T_d(A) - A^d\| \leq C_d$ for every self-adjoint contraction A , with equality for $A = c_d^* I$. This applies to reversible Markov operators on $L^2(\pi)$, where $T_d(P)$ describes discrete-wave (quantum-walk) dynamics and P^d the d -step random walk [1]; equality holds for the symmetric two-state chain with eigenvalues $\{1, c_d^*\}$. Likewise $C_d = \sup_{|z|=1} |q(z^d) - q(z)^d|$ for $q(z) = (z + z^{-1})/2$, since $q(z^d) = T_d(q(z))$ [17]. These are restatements of Corollary 2.2 and Theorem 2.4, not independent results.

Quantum parity. For commuting local unitaries $V_j = \text{diag}(e^{i\theta}, e^{-i\theta})$ and the Greenberger–Horne–Zeilinger state [13], $\langle \otimes_j V_j \rangle = \cos d\theta$ and $\langle V_j \rangle = \cos \theta$, so the factorization defect in this single state is $|\cos d\theta - \cos^d \theta|$ and equals C_d at $\theta = \theta_d^*$. For Hermitian observables with spectrum $\{\pm 1\}$ the joint spectral law is ± 1 -valued, and the relevant sharp constant is the real one, $1 + (1 - 2/d)^d$ (Theorem 6.1).

Normalization and estimation. The ratio $D_d(X)/C_d$ lies in $[0, 1]$ and vanishes exactly under mutual independence; for $d = 2$ it is the UFDm [6]. It should be used with care: it is not invariant under adding an independent coordinate (which leaves D_d unchanged but replaces C_d by C_{d+1}), it is not invariant under monotone transformations of the marginals, and Gaussian laws never exceed $1/C_d$. Moreover, the supremum over all frequencies cannot be estimated consistently from data: the empirical characteristic function is not uniformly consistent on all of $\mathbb{R}^d$ [3, 5], and the uniform estimator of [6] restricts frequencies accordingly. In fact, if N is even and, for each j , the observed values $x_{1j}, \dots, x_{Nj}$ are linearly independent over $\mathbb{Q}$ (almost surely for i.i.d. samples with absolutely continuous marginals), then the unrestricted empirical supremum equals C_d for every such data set, by Kronecker’s theorem and Corollary 2.2.

Mixing. Corollary 2.2 is an unconditional ceiling for multi-factor covariance inequalities under strong mixing, such as the Volkonskii–Rozanov lemma [19]; sharp mixing constants are treated separately.

10 Open problems

1. Classify all extremizers of the profile/universal theorem for $d \geq 3$ and prove a quantitative stability theorem. The $d = 2$ extremal set is larger.
2. Prove irreducibility of the critical polynomial A_d for every d . A companion paper proves broad infinite families, local factorization theorems and large Galois-group results, but the all- d conjecture remains open [15].
3. Prove or refute global discrete concavity $C_{d+2} - 2C_{d+1} + C_d < 0$ for every $d \geq 2$.
4. Extend the structural restrictions beyond the classes already settled in [14]: finite phase alphabets, further fixed-marginal families, and broader Gaussian/elliptical classes.

Supplementary material

A table of C_d , θ_d^* and c_d^* for $2 \leq d \leq 100$, computation scripts, and the previous independent phase-shift/angular proof of Corollary 2.2 accompany this manuscript. The older proof is retained as an independent verification route, not as the primary proof.

Historical note

The uniform characteristic-function factorization defect and the question of its extremal constants, including the limit 2, go back to an unpublished student research project of the author (1996), originally motivated in part by cryptographic applications. The exact finite- d formula, fixed-mean profile, proofs, continuous refinements and the critical-polynomial formulation in this paper were obtained in 2026.

References

- [1] S. Apers and L. Miclo, Quantum walks, the discrete wave equation and Chebyshev polynomials, arXiv:2402.07809 (2024).
- [2] B. Böttcher, M. Keller-Ressel and R. L. Schilling, Distance multivariate: new dependence measures for random vectors, *Ann. Statist.* 47 (2019) 2757–2789.
- [3] S. Csörgő, Limit behaviour of the empirical characteristic function, *Ann. Probab.* 9 (1981) 130–144.
- [4] S. Csörgő, Testing for independence by the empirical characteristic function, *J. Multivariate Anal.* 16 (1985) 290–299.
- [5] S. Csörgő and V. Totik, On how long interval is the empirical characteristic function uniformly consistent?, *Acta Sci. Math. (Szeged)* 45 (1983) 141–149.
- [6] P. Daniušis, S. Juneja, L. Kuzma and V. Marcinkevičius, Measuring statistical dependence via characteristic function IPM, *Entropy* 27(12) (2025) 1254.
- [7] N. Ebrahimi, G. G. Hamedani, E. S. Soofi and H. Volkmer, A class of models for uncorrelated random variables, *J. Multivariate Anal.* 101 (2010) 1859–1871.
- [8] G. G. Hamedani, Sub-independence: an expository perspective, *Comm. Statist. Theory Methods* 42 (2013) 3615–3638.
- [9] I. A. Ibragimov, Some limit theorems for stationary processes, *Theory Probab. Appl.* 7 (1962) 349–382.
- [10] A. Kankainen, *Consistent Testing of Total Independence Based on the Empirical Characteristic Function*, PhD thesis, University of Jyväskylä, 1995.
- [11] Z. Kukorelly, The piling-up lemma and dependent random variables, in *Cryptography and Coding*, LNCS 1746, Springer, 1999, 186–190.
- [12] M. Matsui, Linear cryptanalysis method for DES cipher, *EUROCRYPT '93*, LNCS 765 (1994) 386–397.
- [13] N. D. Mermin, Extreme quantum entanglement in a superposition of macroscopically distinct states, *Phys. Rev. Lett.* 65 (1990) 1838–1840.
- [14] A. Martinevski, *Two-point extremals for characteristic-function dependence with fixed marginals*, companion manuscript (2026).
- [15] A. Martinevski, *Chebyshev–Lucas critical polynomials: local blocks, irreducibility and Galois groups*, companion manuscript (2026).
- [16] N. Pfister, P. Bühlmann, B. Schölkopf and J. Peters, Kernel-based tests for joint independence, *J. R. Stat. Soc. Ser. B* 80 (2018) 5–31.
- [17] T. J. Rivlin, *Chebyshev Polynomials*, 2nd ed., Wiley, 1990.
- [18] S. M. Schennach, Convolution without independence, *J. Econometrics* 211(1) (2019) 308–318.
- [19] V. A. Volkonskii and Yu. A. Rozanov, Some limit theorems for random functions I, *Theory Probab. Appl.* 4 (1959) 178–197.